

transactions are stored permanently and must be verifiable by anyone at any time — not just the original two parties.

To solve this, Blum *et al* introduced Non-Interactive Zero-Knowledge Proofs (NIZKPs). In a non-interactive proof, the prover creates a single piece of evidence (the proof) that can be verified by anyone, at any time, without any further communication. This is essential for blockchain use, where the proof is stored onchain and must be checked by many different nodes.

The privacy problem in open source blockchains

Public blockchains like Bitcoin and Ethereum are designed to be fully transparent. Every transaction is broadcast to the entire network and recorded permanently. While this openness ensures that no single party can manipulate the record, it creates a serious conflict with the need for financial and data privacy.

Despite the pseudonymous nature of blockchain addresses, research has shown that advanced blockchain analytics can often deanonymize users. By correlating onchain transaction data with offchain information, such as exchange records, IP addresses, or social media activity, analysts can often identify who is behind a wallet address. This means that a user's entire transaction history, which might span years, can be exposed.

For enterprise use cases, this problem is even more serious. Supply chain tracking, healthcare data, financial settlements, and voting systems all require a level of confidentiality that a fully transparent blockchain cannot provide. The challenge is to allow anyone to verify that the rules are being followed -- for example, that no coins are being created out of thin air without revealing the specific details of each transaction.

ZKPs directly address this challenge. They allow a blockchain node to verify that a transaction is valid (for example, that the sender has enough funds) without seeing the actual wallet

balances or transaction amounts.

Types of Zero-Knowledge Proofs used in blockchain

Several different types of ZKP have been developed over the years, each with different tradeoffs between proof size, generation speed, verification speed, and security assumptions. The three most important types for blockchain applications are zkSNARKs, zkSTARKs, and Bulletproofs.

zkSNARKs (ZeroKnowledge Succinct NonInteractive Arguments of Knowledge): zkSNARKs are the most widely used type of ZKP in blockchain today. The key properties of zkSNARKs are captured in their name: they are Zero-Knowledge (no information is revealed), succinct (the proof is very small and fast to verify), non-interactive (no back and forth communication is needed), and an argument of knowledge (the prover must actually know the secret to generate a valid proof).

The main strength of zkSNARKs is that they produce very small proof sizes and can be verified in milliseconds, which is critical for a blockchain where many nodes need to verify proofs quickly. A benchmark study by ElHajj and Roelink (2024) found that zkSNARKs produced the smallest proofs among the three major ZKP types.

However, zkSNARKs have one important weakness: they require a one-time trusted setup. During this setup phase, a set of public parameters (called a Common Reference String, or CRS) must be generated. If the secret randomness used in this setup is ever known by an attacker, they could create fake proofs that appear valid, potentially counterfeiting tokens or stealing funds. This is often called the toxic waste problem.

To reduce this risk, many projects use a MultiParty Computation (MPC) ceremony, where many independent parties each contribute to the setup. As long as even one participant destroys their share of the secret, the system remains safe. Zcash's Sapling upgrade involved 87

participants in such a ceremony.

Key use cases: Zcash (shielded transactions), Ethereum zkrollups (Polygon zkEVM, zkSync), StarkWare **zkSTARKs (ZeroKnowledge Scalable Transparent Arguments of Knowledge):** zkSTARKs were introduced in a 2018 paper by Eli BenSasson, Iddo Bentov, Yinon Horesh, and Michael Riabzev as an alternative to zkSNARKs. The 'T' in STARK stands for 'transparent', meaning all the parameters are publicly verifiable and no secret ceremony is needed.

Instead of relying on elliptic curve cryptography (as SNARKs do), STARKs use collision-resistant hash functions. This makes them potentially secure against quantum computers, which are expected to eventually be able to break elliptic curve-based systems. This is a significant long-term advantage.

The downside of zkSTARKs is that they produce much larger proofs than SNARKs. The benchmark study by ElHajj and Roelink found that zkSTARKs generated the largest proofs, which means more storage and higher onchain gas costs. However, they were also the fastest in terms of proof generation and verification time.

Key use cases: StarkEx (used by dYdX and Immutable X), StarkNet

Bulletproofs: Bulletproofs were introduced in 2018 by Bünz, Bootle, Boneh, Poelstra, Wuille, and Maxwell. They are a type of non-interactive ZKP that does not require a trusted setup. The name comes from the fact that they are short (like a bullet) — significantly more compact than older proof systems.

Bulletproofs are particularly efficient for range proofs, which are used to prove that a number falls within a certain range (for example, that a transaction amount is positive and not too large) without revealing the exact number. Monero uses Bulletproofs to enable confidential transactions, which hide transaction amounts. Their adoption reduced transaction data size by up to 80%.